Chapter 7

Constrained Optimization: KKT Conditions and Second-Order Tests

Abstract:
This chapter develops local optimality conditions for constrained problems: feasible directions, tangent cones, constraint qualifications, first-order KKT conditions, second-order tests, and representative examples.

7.1 Introduction

Chapter 1 explained first- and second-order optimality conditions for unconstrained problems, and Chapter 4 introduced the convex sets and convex constraint classes that appear in constrained models. The new issue here is that, near a feasible point, not every direction is admissible. Constrained optimization therefore starts with a local question: how can one recognize or certify a minimizer when the feasible set blocks most directions?

7.2 Problem statement

We begin with the standard constrained problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & g_i(\mathbf{x}) \le 0, \qquad i = 1, \dots, m, \\ & h_j(\mathbf{x}) = 0, \qquad j = 1, \dots, p, \end{aligned} \tag{7.1}$$

where

$$f, g_i, h_j \in C^1.$$

7.2.1 Basic definitions

The basic objects attached to (7.1) are the feasible set, the active constraints, and the Lagrangian.

Definition 7.1 (Feasible set)**.** We define the common domain by

$$D := \operatorname{dom} f \cap \bigcap_{i=1}^{m} \operatorname{dom} g_i \cap \bigcap_{j=1}^{p} \operatorname{dom} h_j$$

and the feasible set by

$$F := \{\mathbf{x} \in D : g_i(\mathbf{x}) \leq 0 \ \forall i, \quad h_j(\mathbf{x}) = 0 \ \forall j\}.$$

Definition 7.2 (Active constraints)**.** Following the notation of this chapter, an inequality constraint is *active* (or *binding*) at $\mathbf{x}$ if it holds with equality. We denote the set of active constraints at $\mathbf{x}$ by

$$\operatorname{Active}(\mathbf{x}) := \{g_i : g_i(\mathbf{x}) = 0, \ i = 1, \ldots, m\} \cup \{h_j : j = 1, \ldots, p\}.$$

Thus the active constraints are precisely the active inequalities together with all equality constraints. When active indices are meant, we use the same notation $\operatorname{Active}(\mathbf{x})$.

Definition 7.3 (Lagrangian)**.** The associated Lagrangian is

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^{p} \mu_j h_j(\mathbf{x}). \tag{7.2}$$

7.2.2 Equivalent descriptions of the same feasible set

Before stating KKT, one modeling issue matters: the feasible set alone does not determine the form of the optimality conditions. What also matters is how that set is represented through equalities and inequalities. The same feasible set may admit different descriptions. For example,

$$h(\mathbf{x}) = 0 \iff \widehat{h}(\mathbf{x}) := h(\mathbf{x})^2 = 0$$

and also

$$h(\mathbf{x}) = 0 \iff \begin{cases} g_1(\mathbf{x}) := h(\mathbf{x}) \leq 0, \\ g_2(\mathbf{x}) := -h(\mathbf{x}) \leq 0. \end{cases}$$

This freedom is useful, but it is also dangerous: some reformulations destroy the constraint qualifications needed for KKT. We will see an explicit example of this failure later in the chapter.

So, KKT is attached to a *constraint representation*, not only to the feasible set itself. Two formulations may define the same set F and still behave differently from the viewpoint of regularity, because they produce different gradients for the active constraints. When we apply KKT, we must therefore work with a representation whose constraints satisfy the required qualification. Some constrained problems can also be reduced to unconstrained ones; in this chapter, however, we keep the constrained formulation and derive the optimality conditions directly from that representation.

7.3 First-order KKT conditions

Theorem 7.1 (First-order KKT conditions)**.** Assume that $f, g_i, h_j \in C^1$, let $\mathbf{x}^\star$ be a local minimizer of (7.1), and assume that a constraint qualification holds at $\mathbf{x}^\star$ (the relevant constraint qualifications are listed in Section 7.3.3). Then $\exists\, \boldsymbol{\lambda}^\star \in \mathbb{R}^m$ and $\exists\, \boldsymbol{\mu}^\star \in \mathbb{R}^p$ such that

1. $$\nabla_{\mathbf{x}} L(\mathbf{x}^\star, \boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star) = \nabla f(\mathbf{x}^\star) + \sum_{i=1}^{m} \lambda_i^\star \nabla g_i(\mathbf{x}^\star) + \sum_{j=1}^{p} \mu_j^\star \nabla h_j(\mathbf{x}^\star) = \mathbf{0}; \tag{7.3}$$
2. $\mathbf{x}^\star \in F$;
3. $\lambda_i^\star \geq 0\ \forall i$;
4. $\lambda_i^\star g_i(\mathbf{x}^\star) = 0\ \forall i$.

Condition (7.3) is the stationarity of the Lagrangian, the second condition is primal feasibility, the third is dual feasibility, and the fourth condition is complementary slackness.

The theorem is the main practical message. The rest of this section derives it from the geometry of feasible directions, the role of constraint qualifications, and the Farkas alternative.

7.3.1 Feasible curves and the tangent cone

Suppose $\mathbf{x}^\star \in F$. If $\mathbf{x}^\star$ is an interior point of F, then locally the problem behaves like an unconstrained one, so we should again expect $\nabla f(\mathbf{x}^\star) = \mathbf{0}$. The interesting case is $\mathbf{x}^\star \in \partial F$, where only some directions remain feasible.

To make that precise, consider all smooth feasible curves issuing from $\mathbf{x}^\star$:

$$S(\mathbf{x}^\star) := \left\{ \mathbf{x}(\cdot) \in C^1 : \mathbf{x}(0) = \mathbf{x}^\star,\ \mathbf{x}(t) \in F \text{ for } t \in [0, t_{\max}] \right\}.$$

Definition 7.4 (Tangent cone)**.** The tangent cone to F at $\mathbf{x}^\star$ is

$$T_F(\mathbf{x}^\star) := \{ \mathbf{d} \in \mathbb{R}^n : \exists\, \mathbf{x}(\cdot) \in S(\mathbf{x}^\star) \text{ with } \dot{\mathbf{x}}(0) = \mathbf{d} \}.$$

This is the collection of all limiting directions of feasible trajectories passing through $\mathbf{x}^\star$; see Figure 7.1.

Claim 7.1 (Tangent-cone necessary condition)**.** If $\mathbf{x}^\star \in F$ is a local minimizer of (7.1), then

$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} \geq 0 \qquad \forall \mathbf{d} \in T_F(\mathbf{x}^\star). \tag{7.4}$$

Proof: Take any feasible curve $\mathbf{x}(\cdot) \in S(\mathbf{x}^\star)$. Since $\mathbf{x}^\star$ is a local minimizer, the one-variable function $t \mapsto f(\mathbf{x}(t))$ has a local minimum at $t = 0$, hence

$$\frac{d}{dt} f(\mathbf{x}(t)) \bigg|_{t=0} \geq 0.$$

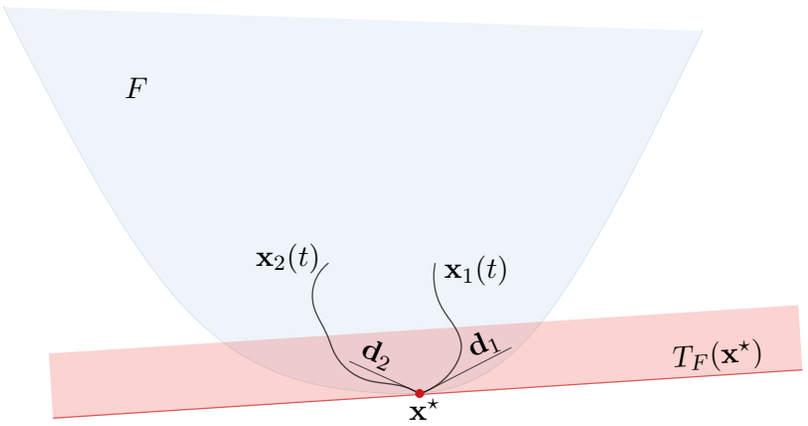

Figure 7.1: Smooth feasible curves $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ issuing from $\mathbf{x}^\star \in F$. Their initial velocities $\mathbf{d}_1 = \dot{\mathbf{x}}_1(0)$ and $\mathbf{d}_2 = \dot{\mathbf{x}}_2(0)$ are feasible first-order directions, hence they belong to the tangent cone $T_F(\mathbf{x}^\star)$.

By the chain rule,

$$\frac{d}{dt} f(\mathbf{x}(t))\bigg|_{t=0} = \nabla f(\mathbf{x}^\star)^\top \dot{\mathbf{x}}(0).$$

Now set $\mathbf{d} = \dot{\mathbf{x}}(0)$. By definition, $\mathbf{d} \in T_F(\mathbf{x}^\star)$, and (7.4) follows. □

The meaning of (7.4) is straightforward: along every feasible first-order direction, the objective must not decrease.

If $\mathbf{x}^\star \in \operatorname{int} F$, then $T_F(\mathbf{x}^\star) = \mathbb{R}^n$, so (7.4) reduces to the unconstrained condition

$$\nabla f(\mathbf{x}^\star) = \mathbf{0}.$$

So far, however, (7.4) is not very convenient to check directly. The purpose of KKT is to rewrite it in terms of the gradients of the constraint functions themselves.

7.3.2 Case 1: one equality constraint

Consider first the problem

$$f(\mathbf{x}) \to \min_{\mathbf{x}}, \qquad h(\mathbf{x}) = 0.$$

At a feasible point $\mathbf{x}^\star$ with $\nabla h(\mathbf{x}^\star) \neq \mathbf{0}$, the tangent cone is

$$T_F(\mathbf{x}^\star) = \left\{ \mathbf{d} \in \mathbb{R}^n : \nabla h(\mathbf{x}^\star)^\top \mathbf{d} = 0 \right\}. \tag{7.5}$$

This geometry is shown in the left panel of Figure 7.2. Indeed, first-order feasibility means that along every feasible curve the constraint stays constant, so its derivative at $t = 0$ must vanish.

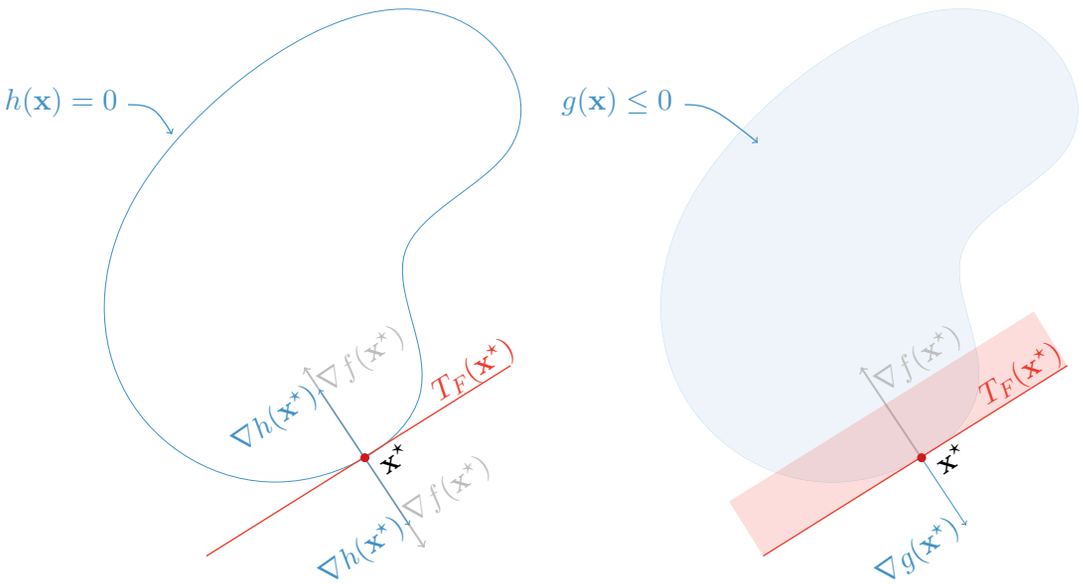

Figure 7.2: Geometry of the first-order KKT conditions in the two basic cases. The gradients $\nabla h(\mathbf{x}^\star)$ and $\nabla g(\mathbf{x}^\star)$ are normal to their level sets because, along a level set, the constraint is constant and therefore the directional derivative is zero. Left: for $h(\mathbf{x}) = 0$, feasible directions satisfy $\nabla h(\mathbf{x}^\star)^\top \mathbf{d} = 0$, so $T_F(\mathbf{x}^\star)$ is the tangent space. Since both $\mathbf{d}$ and $-\mathbf{d}$ are feasible, first-order optimality forces $\nabla f(\mathbf{x}^\star)$ to have no tangent component: $\nabla f(\mathbf{x}^\star) = -\mu \nabla h(\mathbf{x}^\star)$, with $\mu \in \mathbb{R}$ free in sign. Right: for $g(\mathbf{x}) \leq 0$, $\nabla g(\mathbf{x}^\star)$ points toward increasing g, hence outward from the feasible side $g < 0$. Feasible directions satisfy $\nabla g(\mathbf{x}^\star)^\top \mathbf{d} \leq 0$, and the condition $\nabla f(\mathbf{x}^\star)^\top \mathbf{d} \geq 0 \; \forall \mathbf{d} \in T_F(\mathbf{x}^\star)$ forces $\nabla f(\mathbf{x}^\star) = -\lambda \nabla g(\mathbf{x}^\star)$ with $\lambda \geq 0$.

Now apply (7.4). Since $\mathbf{d} \in T_F(\mathbf{x}^\star)$ implies $-\mathbf{d} \in T_F(\mathbf{x}^\star)$, the inequality

$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} \geq 0 \qquad \forall \mathbf{d} \in T_F(\mathbf{x}^\star)$$

forces equality:

$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} = 0 \qquad \forall \mathbf{d} \in T_F(\mathbf{x}^\star).$$

So $\nabla f(\mathbf{x}^\star)$ is orthogonal to the tangent space (7.5). The normal space is spanned by $\nabla h(\mathbf{x}^\star)$, hence $\exists\, \mu \in \mathbb{R}$ such that

$$\nabla f(\mathbf{x}^\star) + \mu \nabla h(\mathbf{x}^\star) = \mathbf{0}.$$

This is exactly the equality-constrained form of KKT stationarity.

Two edge cases are worth keeping in mind.

- If $\nabla f(\mathbf{x}^\star) = \mathbf{0}$, then the condition collapses to the unconstrained one from Chapter 1.
- If $\nabla h(\mathbf{x}^\star) = \mathbf{0}$, then the linearized constraint becomes trivial and imposes no restriction on the direction $\mathbf{d}$. As a consequence, the first-order approximation suggests that every direction is feasible, whereas the actual feasible set may admit only a much smaller set of directions. This is the first sign that a constraint qualification is needed.

7.3.3 Constraint qualifications

Constraint qualifications are introduced precisely to rule out such degenerate situations. Informally, a constraint qualification says that near the current point, the local feasible geometry is captured correctly by the linearized active constraints, namely the constraints in $\mathrm{Active}(\mathbf{x})$.

Definition 7.5 (Common constraint qualifications)**.** At a feasible point $\mathbf{x}^\star \in F$, each of the following is a standard constraint qualification.

1. **LCQ (linear constraint qualification).** All constraint functions are affine with nonzero linear part.
2. **LICQ (linear independence constraint qualification).** The family
$$\{\nabla g_i(\mathbf{x}^\star) : i \in \mathrm{Active}(\mathbf{x}^\star)\} \cup \{\nabla h_j(\mathbf{x}^\star) : j \in \mathrm{Active}(\mathbf{x}^\star)\}$$
is linearly independent.
3. **Slater's condition.** For a convex constrained problem, $\exists\, \bar{\mathbf{x}} \in F$ such that
$$h_j(\bar{\mathbf{x}}) = 0 \quad \forall j, \qquad g_i(\bar{\mathbf{x}}) < 0 \quad \forall i.$$
4. **Weak Slater's condition.** For a convex constrained problem, $\exists\, \bar{\mathbf{x}} \in F$ such that
$$h_j(\bar{\mathbf{x}}) = 0 \quad \forall j,$$
and for the inequalities,
$$g_i(\bar{\mathbf{x}}) \le 0 \text{ for affine } g_i, \qquad g_i(\bar{\mathbf{x}}) < 0 \text{ for the remaining } g_i.$$

The convexity assumptions in the Slater-type conditions refer back to the material of Chapter 4: in the standard convex setting the functions g_i are convex and the equalities h_j are affine.

7.3.4 Case 2: one inequality constraint

Now consider
$$f(\mathbf{x}) \to \min_{\mathbf{x}}, \qquad g(\mathbf{x}) \le 0.$$
If $\mathbf{x}^\star \in \operatorname{int} F$, we are back in the unconstrained case and $\nabla f(\mathbf{x}^\star) = \mathbf{0}$. So we are more interested in a boundary point with $g(\mathbf{x}^\star) = 0$. We know that $\nabla g(\mathbf{x})$ will be orthogonal to the level set and directed outward because, inside the blue region (see Figure 7.2), the value of g is negative, on the boundary it is zero, and the gradient is always directed toward increasing values of the function. Then see the right panel of Figure 7.2:
$$T_F(\mathbf{x}^\star) = \left\{\mathbf{d} \in \mathbb{R}^n : \nabla g(\mathbf{x}^\star)^\top \mathbf{d} \le 0\right\}. \tag{7.6}$$

The tangent-cone condition
$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} \ge 0 \qquad \forall \mathbf{d} \in T_F(\mathbf{x}^\star)$$

requires $\nabla f(\mathbf{x}^\star)$ to have only one possible direction, like the one shown in the right panel of Figure 7.2. Equivalently, $\exists\, \lambda \geq 0$ such that

$$\nabla f(\mathbf{x}^\star) + \lambda \nabla g(\mathbf{x}^\star) = \mathbf{0}.$$

This is the one-constraint inequality version of KKT. As in the case 1 the same notes about constraint qualifications can be made here.

The fourth KKT condition,

$$\lambda g(\mathbf{x}^\star) = 0,$$

just covers all possible cases, which can be separated into two cases:

- if $\lambda = 0$, then ($\mathbf{x}^\star \in \operatorname{int} F$ if $g(\mathbf{x}^\star) < 0$ or $\mathbf{x}^\star \in \partial F$ if $g(\mathbf{x}^\star) = 0$) and $\nabla f(\mathbf{x}^\star) = \mathbf{0}$;
- if $g(\mathbf{x}^\star) = 0$, then $\mathbf{x}^\star \in \partial F$.

Therefore, it forbids cases when $\lambda > 0$ and $g(\mathbf{x}^\star) < 0$, so that if we have inner point $\mathbf{x}^\star$, $\nabla f(\mathbf{x}^\star)$ must be always zero.

7.3.5 Case 3: two inequality constraints

Now consider the next case,

$$f(\mathbf{x}) \to \min_{\mathbf{x}}, \qquad g_1(\mathbf{x}) \leq 0, \qquad g_2(\mathbf{x}) \leq 0,$$

and suppose that both constraints are active at $\mathbf{x}^\star$:

$$g_1(\mathbf{x}^\star) = 0, \qquad g_2(\mathbf{x}^\star) = 0.$$

Let $F_i := \{\mathbf{x} : g_i(\mathbf{x}) \leq 0\}$ for $i = 1, 2$. The two individual linearized tangent cones are

$$T_{F_i}(\mathbf{x}^\star) = \left\{\mathbf{d} \in \mathbb{R}^n : \nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} \leq 0\right\}, \qquad i = 1, 2.$$

Since the feasible set is $F = F_1 \cap F_2$, a direction is feasible to first order only if it satisfies both linearized inequalities. Thus

$$T_F(\mathbf{x}^\star) = T_{F_1}(\mathbf{x}^\star) \cap T_{F_2}(\mathbf{x}^\star) = \left\{\mathbf{d} \in \mathbb{R}^n : \nabla g_1(\mathbf{x}^\star)^\top \mathbf{d} \leq 0,\ \nabla g_2(\mathbf{x}^\star)^\top \mathbf{d} \leq 0\right\}.$$

The tangent-cone condition

$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} \geq 0 \qquad \forall \mathbf{d} \in T_F(\mathbf{x}^\star)$$

therefore restricts $\nabla f(\mathbf{x}^\star)$ to the grey cone in Figure 7.3. In other words,

$$\exists\, \lambda_1, \lambda_2 \geq 0 : \qquad \nabla f(\mathbf{x}^\star) = -\lambda_1 \nabla g_1(\mathbf{x}^\star) - \lambda_2 \nabla g_2(\mathbf{x}^\star).$$

Equivalently,

$$\nabla f(\mathbf{x}^\star) + \lambda_1 \nabla g_1(\mathbf{x}^\star) + \lambda_2 \nabla g_2(\mathbf{x}^\star) = \mathbf{0}, \qquad \lambda_1, \lambda_2 \geq 0.$$

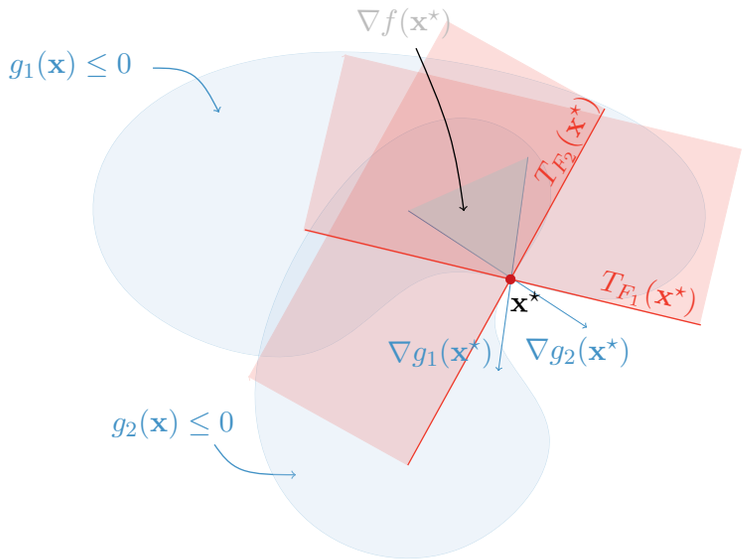

Figure 7.3: Two active inequalities. Each active constraint gives one tangent half-space, $T_{F_i}(\mathbf{x}^\star) = \{\mathbf{d} : \nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} \leq 0\}$, and the final tangent cone is their intersection $T_F(\mathbf{x}^\star) = T_{F_1}(\mathbf{x}^\star) \cap T_{F_2}(\mathbf{x}^\star)$. The grey cone is the set of vectors having nonnegative inner product with every direction in this intersection: $\operatorname{cone}\{-\nabla g_1(\mathbf{x}^\star), -\nabla g_2(\mathbf{x}^\star)\}$. By Claim 7.1, $\nabla f(\mathbf{x}^\star)$ must lie in this grey cone; outside it, the inequality $\nabla f(\mathbf{x}^\star)^\top \mathbf{d} \geq 0 \; \forall \mathbf{d} \in T_F(\mathbf{x}^\star)$ fails.

7.3.6 The general case and Farkas' lemma

Once a constraint qualification holds, the tangent cone at a feasible point $\mathbf{x}^\star$ admits the linear description suggested by the previous cases:

$$T_F(\mathbf{x}^\star) = \left\{ \mathbf{d} \in \mathbb{R}^n : \begin{array}{ll} \nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} \leq 0 & \forall i \in \operatorname{Active}(\mathbf{x}^\star) \\ \nabla h_j(\mathbf{x}^\star)^\top \mathbf{d} = 0 & \forall j \end{array} \right\}. \tag{7.7}$$

So the first-order condition (7.4) becomes a statement about one vector, $\nabla f(\mathbf{x}^\star)$, against a cone described by finitely many linear inequalities and equalities. This is exactly the setting where the cone version of Farkas' lemma applies.

Lemma 7.1 (Farkas)**.** Let

$$K := \{B\mathbf{y} + C\mathbf{w} : \mathbf{y} \geq \mathbf{0}\},$$

where $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{n \times p}$, $\mathbf{y} \in \mathbb{R}^m$, and $\mathbf{w} \in \mathbb{R}^p$. Then for every $\mathbf{g} \in \mathbb{R}^n$, exactly one of the following alternatives holds:

1. $\mathbf{g} \in K$ or
2. $\exists\, \mathbf{d} \in \mathbb{R}^n$ such that

$$\mathbf{g}^\top \mathbf{d} < 0, \qquad B^\top \mathbf{d} \geq \mathbf{0}, \qquad C^\top \mathbf{d} = \mathbf{0}.$$

Geometrically, this lemma is a separation statement for a point and a convex cone: either the point already lies in the cone, or one can separate them by a direction.

To use it, introduce the cone generated by the active constraint normals:

$$N(\mathbf{x}^\star) := \left\{ -\sum_{i \in \mathrm{Active}(\mathbf{x}^\star)} \lambda_i \nabla g_i(\mathbf{x}^\star) - \sum_{j=1}^{p} \mu_j \nabla h_j(\mathbf{x}^\star) : \lambda_i \geq 0 \right\}. \tag{7.8}$$

Applying Lemma 7.1 to $\mathbf{g} = \nabla f(\mathbf{x}^\star)$ and the linear system (7.7) gives two possibilities:

1. $\nabla f(\mathbf{x}^\star) \in N(\mathbf{x}^\star)$ or
2. $\exists\, \mathbf{d} \in \mathbb{R}^n$ such that
$$\begin{aligned} &\nabla f(\mathbf{x}^\star)^\top \mathbf{d} < 0, \\ &\nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} \leq 0 \quad \forall i \in \mathrm{Active}(\mathbf{x}^\star), \\ &\nabla h_j(\mathbf{x}^\star)^\top \mathbf{d} = 0 \quad \forall j. \end{aligned}$$

But the second alternative gives $\mathbf{d} \in T_F(\mathbf{x}^\star)$ with $\nabla f(\mathbf{x}^\star)^\top \mathbf{d} < 0$, contradicting Claim 7.1. Hence only the first alternative can occur.

This finishes a sketch of the proof for theorem 7.1.

7.4 Second-order KKT conditions

If the stronger inequality

$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} > 0 \qquad \forall \mathbf{d} \in T_F(\mathbf{x}^\star) \setminus \{\mathbf{0}\}$$

holds, then every nontrivial feasible first-order motion increases the objective, so $\mathbf{x}^\star$ is already a local minimizer. The subtle case is when the first-order term vanishes in some feasible directions. We then pass to a second-order test.

Assume that first-order KKT holds at $(\mathbf{x}^\star, \boldsymbol{\lambda}, \boldsymbol{\mu})$. For every $\mathbf{d} \in T_F(\mathbf{x}^\star)$,

$$\nabla f(\mathbf{x}^\star)^\top \mathbf{d} = -\sum_{i=1}^{m} \lambda_i \nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} - \sum_{j=1}^{p} \mu_j \nabla h_j(\mathbf{x}^\star)^\top \mathbf{d}.$$

Since $\mathbf{d} \in T_F(\mathbf{x}^\star)$, the equality terms vanish and the inequality terms are nonnegative after multiplication by $-\lambda_i$:

$$-\lambda_i \nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} \geq 0.$$

Therefore a direction with $\nabla f(\mathbf{x}^\star)^\top \mathbf{d} = 0$ must satisfy

$$\nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} = 0 \qquad \forall i \in \mathrm{Active}(\mathbf{x}^\star) \text{ with } \lambda_i > 0.$$

This leads to the critical cone.

Definition 7.6 (Critical cone)**.** Given a feasible point $\mathbf{x}^\star$ and multipliers $\boldsymbol{\lambda} \geq \mathbf{0}$, the critical cone is

$$C(\mathbf{x}^\star, \boldsymbol{\lambda}) := \left\{ \mathbf{d} \in T_F(\mathbf{x}^\star) : \ \nabla g_i(\mathbf{x}^\star)^\top \mathbf{d} = 0 \quad \forall i \in \mathrm{Active}(\mathbf{x}^\star) \text{ and } \lambda_i > 0 \ \right\}. \tag{7.9}$$

Theorem 7.2 (Second-order KKT necessity)**.** Assume that $f, g_i, h_j \in C^2$, let $\mathbf{x}^\star$ be a local minimizer of (7.1), and assume that a constraint qualification holds at $\mathbf{x}^\star$. Then $\exists\,(\boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star)$ satisfying the first-order KKT conditions from Theorem 7.1 and, in addition,

$$\mathbf{d}^\top \nabla^2_{\mathbf{xx}} L(\mathbf{x}^\star, \boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star)\mathbf{d} \geq 0 \qquad \forall \mathbf{d} \in C(\mathbf{x}^\star, \boldsymbol{\lambda}^\star). \tag{7.10}$$

Theorem 7.3 (Second-order KKT sufficiency)**.** Assume that $f, g_i, h_j \in C^2$, that a constraint qualification holds at the feasible point $\mathbf{x}^\star$, and that $(\boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star)$ satisfies the first-order KKT conditions. If, in addition,

$$\mathbf{d}^\top \nabla^2_{\mathbf{xx}} L(\mathbf{x}^\star, \boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star)\mathbf{d} > 0 \qquad \forall \mathbf{d} \in C(\mathbf{x}^\star, \boldsymbol{\lambda}^\star) \setminus \{\mathbf{0}\}, \tag{7.11}$$

then $\mathbf{x}^\star$ is a local minimizer.

The point of (7.10) and (7.11) is that the Hessian of the Lagrangian is not checked on the whole space $\mathbb{R}^n$, but only on the critical directions left undecided by first-order KKT. However, sometimes we check the Hessian inequality on the whole space, because there are cases when it is hard to explicitly construct the critical cone.

7.5 Worked examples

7.5.1 A closed half-line

Example 7.1 (Minimizing on $x \geq 0$)**.** Consider

$$\min_x x^2 \qquad \text{subject to } x \geq 0.$$

To fit the standard form $g(x) \leq 0$, write the constraint as $-x \leq 0$. Then

$$L(x, \lambda) = x^2 - \lambda x.$$

The KKT system is

$$2x - \lambda = 0, \qquad x \geq 0, \qquad \lambda \geq 0, \qquad \lambda x = 0.$$

There are two cases.

- If $\lambda = 0$, then $2x = 0$, so $x = 0$.
- If $\lambda > 0$, then complementary slackness gives $x = 0$, and then $2x - \lambda = 0$ forces $\lambda = 0$, a contradiction.

So the only KKT point is $(x^\star, \lambda^\star) = (0, 0)$. Moreover,

$$\nabla^2_{xx} L(x, \lambda) = 2 > 0,$$

so the second-order sufficient condition holds and $x^\star = 0$ is a local minimum.

7.5.2 An open feasible set

Example 7.2 (The infimum is not attained on $x > 0$)**.** Consider

$$\min_x x^2 \qquad \text{subject to } x > 0.$$

The feasible set is open, so every feasible point is an interior point. Hence the necessary condition for a local minimizer is the unconstrained one:

$$\frac{d}{dx}(x^2) = 2x = 0.$$

Its only solution is $x = 0$, but $x = 0$ is infeasible. Therefore the problem has no minimizer. The infimum equals 0, but it is approached only as $x \downarrow 0$.

7.5.3 A true minimizer where KKT fails because the constraint qualification fails

Example 7.3 (Why a constraint qualification matters)**.** Consider

$$\min_x x \qquad \text{subject to } x^2 \leq 0.$$

The feasible set consists of the single point $x = 0$, so $x^\star = 0$ is the unique minimizer. Now test KKT. The Lagrangian is

$$L(x, \lambda) = x + \lambda x^2.$$

The KKT system becomes

$$1 + 2\lambda x = 0, \qquad x^2 \leq 0, \qquad \lambda \geq 0, \qquad \lambda x^2 = 0.$$

Again split into two cases.

- If $\lambda = 0$, then the stationarity equation becomes $1 = 0$, which is impossible.
- If $\lambda > 0$, then complementary slackness yields $x = 0$, and the stationarity equation again becomes $1 = 0$.

So KKT has no solution, even though the problem has a minimizer. This is not a contradiction: the constraint is not regular at the minimizer because

$$\nabla(x^2)\big|_{x=0} = 0.$$

This is exactly the kind of bad reformulation warned about at the beginning of the chapter.

7.5.4 A quadratic on the unit disk

Example 7.4 (Second-order KKT distinguishes true minima from a false one)**.** Consider

$$\min_{x,y} \; x^2 - y^2 \qquad \text{subject to } x^2 + y^2 \leq 1.$$

Slater's condition holds: for example, $(0,0)$ is strictly feasible since $0^2 + 0^2 - 1 < 0$. Therefore the KKT conditions are applicable. The Lagrangian is

$$L(x, y, \lambda) = x^2 - y^2 + \lambda(x^2 + y^2 - 1).$$

The KKT system is

$$\begin{aligned} 2x + 2\lambda x &= 2x(1+\lambda) = 0, \\ -2y + 2\lambda y &= 2y(\lambda - 1) = 0, \\ x^2 + y^2 &\leq 1, \\ \lambda &\geq 0, \\ \lambda(x^2 + y^2 - 1) &= 0. \end{aligned} \tag{7.12}$$

First solve (7.12).

- If $\lambda = 0$, then $x = 0$ and $y = 0$, so $(0,0)$ satisfies KKT.
- If $\lambda > 0$, then complementary slackness gives $x^2 + y^2 = 1$. The first two equations imply $x = 0$ and $\lambda = 1$, hence $y = \pm 1$. So $(0,1)$ and $(0,-1)$ are also KKT points.

Now use the second-order test. The Hessian of the Lagrangian is

$$\nabla^2_{\mathbf{xx}} L(x, y, \lambda) = \begin{bmatrix} 2(1+\lambda) & 0 \\ 0 & 2(\lambda - 1) \end{bmatrix}.$$

At $(0,0)$ we have $\lambda = 0$, so

$$\nabla^2_{\mathbf{xx}} L(0,0,0) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Here the constraint is inactive, so the tangent cone is all of $\mathbb{R}^2$, and the critical cone is also $\mathbb{R}^2$. The Hessian is indefinite on $\mathbb{R}^2$, therefore $(0,0)$ is *not* a local minimum.

At $(0,1)$ we have $\lambda = 1$, hence

$$\nabla^2_{\mathbf{xx}} L(0,1,1) = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}.$$

The active constraint gradient is

$$\nabla g(0,1) = \nabla(x^2 + y^2 - 1)\big|_{(0,1)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix},$$

so

$$T_F(0,1) = \left\{ \mathbf{d} = (d_1, d_2)^\top : 0 \cdot d_1 + 2d_2 \leq 0 \right\}.$$

Because $\lambda = 1 > 0$, the critical cone imposes equality:

$$C((0,1),1) = \{\mathbf{d} \in T_F(0,1) : 2d_2 = 0\} = \{\mathbf{d} : d_2 = 0\}.$$

Therefore, $\forall \mathbf{d} \in C((0,1),1) \setminus \{\mathbf{0}\}$,

$$\mathbf{d}^\top \nabla^2_{\mathbf{xx}} L(0,1,1)\mathbf{d} = 4d_1^2 > 0.$$

So $(0,1)$ is a local minimum.

Exactly the same argument applies to $(0,-1)$, where

$$T_F(0,-1) = \left\{\mathbf{d} = (d_1,d_2)^\top : -2d_2 \le 0\right\}, \qquad C((0,-1),1) = \{\mathbf{d} : d_2 = 0\},$$

and again the quadratic form equals $4d_1^2 > 0$ on nonzero critical directions. Hence $(0,-1)$ is also a local minimum.

7.5.5 A determinant maximization example

Example 7.5 (Maximizing $\det(\mathbf{X})$ under one trace constraint)**.** Recall from Chapter 4 that $\mathbb{S}^n_{++}$ denotes the cone of symmetric positive definite matrices. Let $\mathbf{A} \in \mathbb{S}^n_{++}$ and $b > 0$, and consider

$$\max_{\mathbf{X}\in\mathbb{S}^n_{++}} \det(\mathbf{X}) \qquad \text{subject to } \langle \mathbf{A}, \mathbf{X}\rangle \le b,$$

where

$$\langle \mathbf{A}, \mathbf{X}\rangle := \operatorname{tr}(\mathbf{A}^\top \mathbf{X}).$$

To fit the chapter's minimization form, rewrite the problem as

$$\min_{\mathbf{X}\in\mathbb{S}^n_{++}} -\det(\mathbf{X}) \qquad \text{subject to } \operatorname{tr}(\mathbf{A}^\top \mathbf{X}) - b \le 0.$$

The constraint qualification holds because the inequality constraint is affine in $\mathbf{X}$. The Lagrangian is

$$L(\mathbf{X},\lambda) = -\det(\mathbf{X}) + \lambda\big(\operatorname{tr}(\mathbf{A}^\top \mathbf{X}) - b\big).$$

For $\mathbf{X} \in \mathbb{S}^n_{++}$, the differential of the determinant is

$$d(\det(\mathbf{X})) = \det(\mathbf{X})\operatorname{tr}(\mathbf{X}^{-1} d\mathbf{X}),$$

so

$$\begin{aligned} dL &= -\det(\mathbf{X})\operatorname{tr}(\mathbf{X}^{-1} d\mathbf{X}) + \lambda \operatorname{tr}(\mathbf{A}^\top d\mathbf{X}) \\ &= \operatorname{tr}\Big(\big(-\det(\mathbf{X})\mathbf{X}^{-1} + \lambda \mathbf{A}\big) d\mathbf{X}\Big), \end{aligned}$$

because $\mathbf{X}$ and $\mathbf{A}$ are symmetric. Hence the KKT system is

$$\begin{aligned} -\det(\mathbf{X})\mathbf{X}^{-1} + \lambda \mathbf{A} &= \mathbf{0}, \\ \operatorname{tr}(\mathbf{A}^\top \mathbf{X}) &\le b, \\ \lambda &\ge 0, \\ \lambda\big(\operatorname{tr}(\mathbf{A}^\top \mathbf{X}) - b\big) &= 0. \end{aligned} \tag{7.13}$$

Now solve (7.13). If $\lambda = 0$, then the first equation gives

$$\det(\mathbf{X})\mathbf{X}^{-1} = \mathbf{0},$$

which is impossible because $\mathbf{X} \succ 0$ implies $\det(\mathbf{X}) > 0$ and $\mathbf{X}$ is invertible. Therefore $\lambda > 0$, and complementary slackness forces the constraint to be active:

$$\operatorname{tr}(\mathbf{A}^\top \mathbf{X}) = b.$$

From stationarity we obtain

$$\mathbf{X} = \frac{\det(\mathbf{X})}{\lambda}\mathbf{A}^{-1} =: \alpha \mathbf{A}^{-1}.$$

Substituting this into the active constraint gives

$$b = \operatorname{tr}(\mathbf{A}^\top \mathbf{X}) = \alpha \operatorname{tr}(\mathbf{A}^\top \mathbf{A}^{-1}) = \alpha \operatorname{tr}(\mathbf{I}) = \alpha n.$$

Hence

$$\alpha = \frac{b}{n}, \qquad \mathbf{X}^\star = \frac{b}{n}\mathbf{A}^{-1}.$$

This matrix is feasible and lies in $\mathbb{S}^n_{++}$, so the optimizer is

$$\boxed{\mathbf{X}^\star = \frac{b}{n}\mathbf{A}^{-1}}.$$

7.5.6 DeepFool as a constrained quadratic subproblem

Example 7.6 (DeepFool adversarial attack)**.** This example shows how the KKT machinery of this chapter also appears in a modern machine-learning application. Let

$$f : \mathbb{R}^n \to \mathbb{R}^K, \qquad \hat{k}(\mathbf{x}) := \arg\max_{1 \le k \le K} f_k(\mathbf{x}),$$

where $f_k(\mathbf{x})$ is the score of class k. For an input $\mathbf{x}$ with predicted class

$$y := \hat{k}(\mathbf{x}),$$

the DeepFool attack seeks a small perturbation $\mathbf{r}$ that changes the predicted label. The basic objective is

$$\min_{\mathbf{r}} \frac{1}{2}\|\mathbf{r}\|_2^2 \qquad \text{subject to } \hat{k}(\mathbf{x} + \mathbf{r}) \neq y.$$

This constraint is nonlinear, so DeepFool replaces it by a first-order model. Linearizing each score around the current point gives

$$f_k(\mathbf{x} + \mathbf{r}) \approx f_k(\mathbf{x}) + \nabla f_k(\mathbf{x})^\top \mathbf{r}.$$

The predicted class changes when at least one competing score catches the current winning score. Locally, we therefore ask how far one must move before the first-order

model of f_j ties the first-order model of f_y. For a fixed competing class $j \neq y$, this linearized boundary between class y and class j is

$$f_y(\mathbf{x}) + \nabla f_y(\mathbf{x})^\top \mathbf{r} = f_j(\mathbf{x}) + \nabla f_j(\mathbf{x})^\top \mathbf{r}.$$

So the local step is the equality-constrained problem

$$\min_{\mathbf{r}} \frac{1}{2}\|\mathbf{r}\|_2^2 \qquad \text{subject to } (\nabla f_y(\mathbf{x}) - \nabla f_j(\mathbf{x}))^\top \mathbf{r} + f_y(\mathbf{x}) - f_j(\mathbf{x}) = 0. \tag{7.14}$$

This is exactly the kind of problem treated earlier in the chapter: a strictly convex quadratic objective with one affine equality constraint. Its Lagrangian is

$$L(\mathbf{r}, \mu) = \frac{1}{2}\|\mathbf{r}\|_2^2 + \mu\Big((\nabla f_y(\mathbf{x}) - \nabla f_j(\mathbf{x}))^\top \mathbf{r} + f_y(\mathbf{x}) - f_j(\mathbf{x})\Big).$$

The KKT conditions are

$$\nabla_{\mathbf{r}} L(\mathbf{r}, \mu) = \mathbf{r} + \mu(\nabla f_y(\mathbf{x}) - \nabla f_j(\mathbf{x})) = \mathbf{0},$$

together with the equality constraint from (7.14). From stationarity,

$$\mathbf{r} = -\mu(\nabla f_y(\mathbf{x}) - \nabla f_j(\mathbf{x})).$$

Substituting this into the constraint yields

$$-\mu\|\nabla f_y(\mathbf{x}) - \nabla f_j(\mathbf{x})\|_2^2 + f_y(\mathbf{x}) - f_j(\mathbf{x}) = 0,$$

so

$$\mu = \frac{f_y(\mathbf{x}) - f_j(\mathbf{x})}{\|\nabla f_y(\mathbf{x}) - \nabla f_j(\mathbf{x})\|_2^2}.$$

Therefore the smallest-norm linearized perturbation that reaches class j is

$$\mathbf{r}_j = -\frac{f_j(\mathbf{x}) - f_y(\mathbf{x})}{\|\nabla f_j(\mathbf{x}) - \nabla f_y(\mathbf{x})\|_2^2}\big(\nabla f_j(\mathbf{x}) - \nabla f_y(\mathbf{x})\big). \tag{7.15}$$

Its norm is

$$\|\mathbf{r}_j\|_2 = \frac{|f_j(\mathbf{x}) - f_y(\mathbf{x})|}{\|\nabla f_j(\mathbf{x}) - \nabla f_y(\mathbf{x})\|_2}.$$

Hence the locally closest class boundary is obtained by choosing

$$\ell(\mathbf{x}) := \arg\min_{j \neq y} \frac{|f_j(\mathbf{x}) - f_y(\mathbf{x})|}{\|\nabla f_j(\mathbf{x}) - \nabla f_y(\mathbf{x})\|_2}. \tag{7.16}$$

The corresponding step is

$$\mathbf{r} = \mathbf{r}_{\ell(\mathbf{x})}.$$

DeepFool then applies this construction iteratively. Starting from $\mathbf{x}_0 = \mathbf{x}$, it keeps updating

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{r}_i$$

as long as the predicted class remains equal to the original label y. At step i, one recomputes the gradients and scores at $\mathbf{x}_i$, selects

$$\ell_i = \arg\min_{j \neq y} \frac{|f_j(\mathbf{x}_i) - f_y(\mathbf{x}_i)|}{\|\nabla f_j(\mathbf{x}_i) - \nabla f_y(\mathbf{x}_i)\|_2},$$

and uses

$$\mathbf{r}_i = -\frac{f_{\ell_i}(\mathbf{x}_i) - f_y(\mathbf{x}_i)}{\|\nabla f_{\ell_i}(\mathbf{x}_i) - \nabla f_y(\mathbf{x}_i)\|_2^2} \big(\nabla f_{\ell_i}(\mathbf{x}_i) - \nabla f_y(\mathbf{x}_i)\big).$$

The final output is the accumulated perturbation, often with a small overshoot factor:

$$\hat{\mathbf{r}} = (1+\eta)(\mathbf{x}_m - \mathbf{x}_0), \qquad \eta \approx 0.02,$$

where m is the first index for which $\hat{k}(\mathbf{x}_m) \neq y$. The overshoot is a practical correction: each computed step is the shortest step to a locally linearized boundary, so without the factor the final point may lie exactly on the approximate boundary or may still remain on the original side of the true nonlinear boundary. Multiplying by $1+\eta$ moves the point slightly farther in the same direction and makes the label change robust to curvature and numerical instabilities.